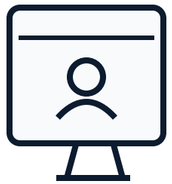
OpenRank Visualization Platform: System Architecture

LLM Open-Source Ecosystem Health Analytics Based on OpenDigger Metrics

1 User & Client



- WEB Web Browser
- USR Student / Analyst / Researcher
- BAR Interactive Dashboard Access

HTTPS Request

Rendered Views

2 Vue 3 Frontend

- V Vite Build System
- RT Vue Router
- P Pinia State Management
- API API Service Layer
- EC ECharts Visualization Engine

Main Views

- H Home Dashboard
- S LLM Ecosystem Summary
- L Visual Lab
- * Favorites
- + Login / Register

Reusable Components

- LN Metric Line Chart
- RD Radar Compare
- DB Contributor Bar Chart

Receives JSON Data from Backend — DOC → CH → Renders Charts & Dashboard Views

REST API Requests (JSON)

JSON Responses (Data, Reports, AI Insights, Favorites)

3 Flask Backend Service

- APP Application Entry: main.py

REST APIs

API OpenDigger API Module	/api/llm/summary
API LLM Summary / Report API	/api/llm/rank/metrics
API Contributor Risk API	/api/llm/report
API Auth API	/api/auth/login
API Auth API (Register)	/api/auth/register
API Favorites API	/api/favorites
API Contributor Risk API	/api/health/contributors/risk

Middleware / Services

- JWT JWT Authentication
- RL API Rate Limiting
- C CORS
- DB Request Cache

Process Requests

Cached Data / DB Results

Query / Read (Results)

4 Data Processing & Storage

Data Processing Pipeline

- DF Data Fetcher
- MU Metric Utilities
- CFG Config Loader
- HS Health Score Calculation
- TS Time-Series Aggregation
- DC Data Cache

Health Score =
0.5 x OpenRank + 0.3 x Activity + 0.2 x Stability

Processed Metrics (Parsed)

SQLite Database

- U User
- * Favorite
- M MetricSeries

Fetch Data (API / Scrape)

5 External Data Sources

- OD OpenDigger
- OR OpenRank Metrics
- GH GitHub Repository Metadata
- PA Project Activity Metrics
- DV Developer / Contributor Metrics

Metric Ingestion

- OpenRank
- Activity
- Stars
- Forks
- Issues
- Pull Requests
- Contributors

Platform Capabilities

- BAR Interactive Visualization
- AI AI Interpretation
- IMG PNG Export
- PDF Report Export
- * Favorites Management
- SH Risk Analysis
- 123 Health Ranking
- PIE Multi-chart Visual Analytics

Legend: Infrastructure / Backend · Data Processing · Analytics / AI Interpretation · Visualization / Capabilities

Primary Data Flow · Response / Return Flow · Bidirectional / Data Access